ADK 2.7.1 ↔ Hermes A2A — Proven Local POC Setup

Status: WORKING POC
Environment: Zorin OS
ADK: 2.7.1
Hermes: Stock Hermes Agent
Protocol: A2A v1.0 / JSON-RPC
Hermes A2A Port: 9900
Topology: Localhost only

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1. Final Working Architecture

```
Browser
  │
  ▼
ADK Dev UI
http://127.0.0.1:8000
  │
  ▼
ADK RemoteA2aAgent
hermes_agent
  │
  │ A2A + Bearer Token
  ▼
Hermes Gateway
http://127.0.0.1:9900
  │
  ▼
Hermes Agent
default profile
  │
  ▼
Configured Model Provider
```

The interactive Hermes CLI does **not** need to remain running.

The Hermes Gateway is the process serving the inbound A2A endpoint.

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2. Upgrade ADK

Original starter-kit version:

```
google-adk==1.13.0
```

Upgrade:

```
pip install --upgrade google-adk
```

Verified version:

```
google-adk==2.7.1
```

Existing starter-kit agents were tested after the upgrade and continued working.

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3. Install ADK A2A Support

`RemoteA2aAgent` exists in ADK, but the A2A SDK is an optional dependency.

Install it while keeping ADK pinned:

```
pip install "google-adk[a2a]==2.7.1"
```

Verify:

```
python -c "from google.adk.agents.remote_a2a_agent import RemoteA2aAgent;
print('A2A OK')"
```

Expected:

```
A2A OK
```

---

4. Enter the Hermes Environment

Hermes was installed at:

```
~/.hermes/hermes-agent
```

Open a normal terminal:

```
cd ~/.hermes/hermes-agent
source venv/bin/activate
```

Verify:

```
which hermes
```

Expected:

```
/home/moose/.hermes/hermes-agent/venv/bin/hermes
```

---

5. Configure Hermes A2A

Run:

```
hermes gateway setup
```

Select:

```
A2A
```

Use:

```
Inbound A2A port: 9900
Agent name: ADK Hermes
```

For the local POC, a peer credential was created:

```
adk:stark-test-001
```

Meaning:

```
Peer identity: adk
Bearer token:  stark-test-001
```

No shared bearer token was configured.

No remote bind host was configured.

The POC therefore remains local to the Zorin machine.

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6. Hermes Setup Quirk Discovered

During setup, pressing Enter on:

```
Inbound A2A port (default 9900):
```

caused Hermes to conceptually use its default port, but it did **not** persist:

```
A2A_PORT=9900
```

The A2A setup menu therefore continued reporting:

```
A2A (not configured)
```

Inspection of:

```
plugins/platforms/a2a/__init__.py
```

showed that Hermes considers A2A connected when either its plugin config is explicitly enabled or the `A2A_PORT` environment variable exists.

The local POC was corrected with:

```
echo 'A2A_PORT=9900' >> ~/.hermes/.env
```

Final verified values:

```
A2A_AGENT_NAME="ADK Hermes"
A2A_PEER_TOKENS=adk:stark-test-001
A2A_PORT=9900
```

For future installations, explicitly enter:

```
9900
```

rather than leaving the port prompt blank.

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7. Start the Hermes A2A Service

From the activated Hermes environment:

```
hermes gateway run
```

Leave this process running.

The interactive Hermes CLI does **not** need to be running.

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8. Verify the Hermes Agent Card

From another terminal:

```
curl http://127.0.0.1:9900/.well-known/agent-card.json
```

The successful POC returned an Agent Card containing, among other fields:

```
name: ADK Hermes
url: http://127.0.0.1:9900/
protocolBinding: JSONRPC
```

```
protocolVersion: 1.0
streaming: true
```

The card also advertised Hermes toolsets/capabilities.

At this point:

```
Hermes A2A Server = VERIFIED
```

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9. Create the ADK Remote Hermes Agent

Add:

```
hermes_agent/
├── __init__.py
└── agent.py
```

`hermes_agent/__init__.py`

```
from . import agent
```

`hermes_agent/agent.py`

```
import httpx

from google.adk.agents.remote_a2a_agent import RemoteA2aAgent


root_agent = RemoteA2aAgent(
    name="hermes_agent",
    description="Local Hermes Agent connected through A2A.",
    agent_card="http://127.0.0.1:9900/.well-known/agent-card.json",
    httpx_client=httpx.AsyncClient(
        headers={
            "Authorization": "Bearer stark-test-001",
        }
    ),
)
```

POC note: The bearer token is hardcoded only for this local experiment.

Do **not** keep credentials in source code for deployment.

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10. Start ADK

From the ADK project virtual environment:

```
adk web
```

Open the ADK Dev UI and select:

```
hermes_agent
```

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11. Proof Test

Send:

```
hello?
```

Then:

```
what is your name?
```

Observed response:

```
I'm Hermes Agent, an AI assistant built by Nous Research.
I'm running under the default profile on this machine.
```

This proved the complete request path:

```
ADK Dev UI
   ↓
RemoteA2aAgent
   ↓
A2A
```

```
   ↓
Hermes Gateway
   ↓
Hermes Agent
   ↓
response
   ↓
ADK Dev UI
```

---

12. POC Verdict

PASS

The experiment proved:

- ADK 2.7.1 can consume a Hermes Agent through A2A.
- Hermes exposes a valid A2A Agent Card.
- Bearer-authenticated ADK requests reach Hermes.
- Hermes responds through its normal agent runtime.
- The response successfully returns into ADK Dev UI.
- The interactive Hermes CLI is not required during A2A operation.
- The Hermes Gateway is the service boundary ADK depends on.

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13. Production Changes Required

The local POC configuration is **not** a production configuration.

Before deployment:

- Remove the bearer token from Python source.
- Generate a strong random peer credential.
- Store the credential in a secret manager.
- Terminate traffic with HTTPS/TLS.
- Do not expose raw port 9900 directly to the public Internet.
- Restrict authenticated peer identities.
- Apply network/firewall restrictions.
- Limit advertised Hermes toolsets.
- Preserve Hermes state on persistent storage.
- Add gateway health monitoring.
- Add A2A audit-log collection.
- Add rate limits and resource limits.
- Run regression tests against ADK after every deployment change.

Known-Good Local Commands

Hermes

```
cd ~/.hermes/hermes-agent
source venv/bin/activate
hermes gateway run
```

Verify Hermes

```
curl http://127.0.0.1:9900/.well-known/agent-card.json
```

ADK

```
cd ~/python/adk-2.x-and-harmes-v1
source .venv/bin/activate
adk web
```

Baseline rule: preserve this working configuration before experimenting with ADK graphs, multiple Hermes agents/profiles, cloud deployment, or expanded A2A orchestration.